

KAI GOODALL

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SKILLS

Design & Simulation: SolidWorks & Fusion 360 (sheet metal, assemblies, drawings), ANSYS (FEA/CFD), MATLAB
Manufacturing: CNC & manual machining, MIG/TIG welding, soldering, metal & polymer 3D printing
Electronics & Software: Python, C++, embedded systems (ESP32), sensor integration, G-code, Git
Test & Validation: Test planning, automated data logging & analysis, reliability testing, root cause analysis

TECHNICAL EXPERIENCE

Reusables | Hardware Engineering Intern

Vancouver, Canada

IoT return bins for reusable containers at university dining halls and sports venues

Jun 2026 – Aug 2026

- **Failure Investigation:** Owned a fleet-wide door-motor failure investigation, designing each test from the last one's data; across 110k logged cycles on 4 test rigs, traced faults to wear-driven motor binding and cut the fault rate from 23% to <1%.
- **Production Impact:** Developed a motor burn-in routine now part of pre-shipment testing for every unit deployed, and shipped motor-unbind firmware running across the production fleet.
- **Motor Selection:** Benchmarked candidate BLDC motors for the 2.0 platform using Python serial-logging scripts, oscilloscopes, and ammeters, selecting the motor now in production.
- **Sensor Integration:** Integrated ToF, rotary position, and inductive proximity sensors into the 2.0 platform, replacing unreliable reed switches; designed their sheet-metal mounting interfaces and set board requirements with the EE contractor.
- **Ethernet Upgrade:** Owned the mechanical design for Ethernet end to end: set requirements, sourced waterproof connectors, iterated designs, and integration-tested samples; revised 10+ sheet-metal parts, fully redesigning 2.

Waterloo Rocketry | Propulsion Subsystem Member

Waterloo, Canada

Team set the amateur liquid-bipropellant altitude world record with Polaris (63,497 ft, 2026)

Sep 2024 – Present

- **Critical Manufacturing:** Machined the injector bulkhead for Liquid Static Fire 4 on a manual lathe, mill, and rotary table; the part withstood a 17 kN peak load, resolving the failure point from the previous test.
- **Test Operations:** Served as primary operator for a 9 kN liquid engine static fire, running 650 psi propellant feed and safety-critical valve sequencing through a full-duration 10 s burn.
- **Materials Selection:** Selected and sourced C18150 (CuCrZr) and C12200 copper alloys for a regeneratively cooled chamber, supplying high-temperature properties for structural FEA and CFD.
- **Brazing R&D:** Developed furnace-brazing methods for coaxial shear injectors with the Propulsion Lead, joining 316L/304L faceplates to hypodermic tubing with silver and brass fillers and sectioning joints to assess bond integrity.

Solo Grants (formerly Merge) | Technical Program Lead

San Francisco, USA (Remote)

Microgrants for individuals building technically ambitious hardware, software, and biotech prototypes

Apr 2024 – Present

- **Technical Validation:** Sole technical evaluator for 2,000+ applications spanning robotics, aerospace, biotech, and chip design, auditing feasibility and build plans; granted \$50k USD to 50+ projects that have since raised \$12M+ USD.
- **Systems & Automation:** Automated review and email workflows with Airtable and Claude Code skills through a 50× volume increase, cutting review time 85% and saving 20–40 hours a month.

Solideon | Additive Manufacturing Engineering Intern

San Francisco, USA

Autonomous, deployable robotic wire-arc additive manufacturing (WAAM) systems

Jun 2024 – Sep 2024

- **Robotic Calibration:** Designed and built a calibration fixture for 6-axis robotic WAAM heads (extrusion stand, CNC-machined faceplate, printed probe), enabling automated sub-mm laser-gate verification and 10× precision.
- **Fixturing:** Designed and built high-rigidity work-holding for an 8 ft × 4 ft × 4 in aluminum customer part, drilling 33 Ø1.5 in bores to restrain it through WAAM thermal loads; the part printed to tolerance without warping.

Telesat | IV&V Systems Engineering Intern

Ottawa, Canada · Jul 2023 – Dec 2023

- **Network Emulation:** Verified routing and latency requirements for a 198-satellite LEO constellation via network emulation.

PROJECTS

3-Axis CNC Mill from a VEX IQ Kit | Software, Integration & Test Lead

Jan 2026 – Apr 2026

- **Machine Build:** On a 4-person team, turned a VEX IQ robotics kit into an 8 × 8 × 5 in CNC mill (rack-and-pinion XY, dual-motor leadscrew Z) that completed full G-code cuts in floral foam; led all integration, testing, and debugging.
- **Software:** Wrote all software in VEX C++: a G-code interpreter converting each line to encoder-counted moves and spindle speed, run in sub-8 KB file chunks to avoid memory crashes, plus motor-current auto-probing to locate the stock.

EDUCATION

University of Waterloo | BAsC, Mechanical Engineering (Co-op)

Waterloo, Canada

President's Scholarship (95%+ admission average) · Dean's List (Winter 2025)

Expected Apr 2030